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PHY 414 — Past Questions & Materials
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Course: PHY 414
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — PHY 414 COMPUTATIONAL PHYSICS Pq

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION

Course Code: PHY 414
Course Title: Computational Physics
Course Unit: 3Units
Session: 2019/2020
Time Allowed: 2 1/2 hours
Date: march 2021
Instruction: You are expected to answer four questions out of the available six.

Question 1

a) The contraction of a spring produced by the load (y kg wt) carried by the spring is tabulated as follows ;

X	5	10	15	20	28
Y	49	105	172	253	352

Use Newtons interpolation method to determine the load that yields the contraction of the spring 7mm and 22mm.

b) Approximate each of these integrals using the Simple Trapezium Rule and work out the error bound to 6 decimal places for the simple trapezium method approximations.

(i) $\int_1^2 e^{-x^2/2} dx$.

(ii) $\int_1^5 \sqrt{x} dx$.

(iii) $\int_1^2 \ln(x) dx$.

(iv) $\int_0^{\pi/4} \sin(x) dx$.

Question 2

a) The tabulated of x and y are as follows;

X	1.0	1.2	1.4	1.6	1.8	2.0	2.2
y	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

Determine the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for $x=1.2$, $x=1.6$ and $x=2.2$.

b) Apply the fourth-order Runge- Kutta method with $h=0.1$ to determine an approximation to the solution to the initial value problem at $x = 1$, $y' = y - x$, $y(0) = 1/2$. (let your n be from 1 -6).

Question 3

a) Using the initial value problem $y' = y - x$, $y(0) = 1/2$, Use Eulers method with $h=0.1$ to obtain an approximation to $y(1)$, Given the exact solution to the initial value problem as

$$y(x) = x + 1 - \frac{1}{2}e^x.$$

Compare the errors in the approximations to $y(1)$. (let n be from 1-5).

- b) Using 4 sub intervals in the composite Simpson rule approximate $\int_0^2 \cosh(x) dx$ and $\int_1^2 \ln(x) dx$.

Question 4

- a) Apply the modified Euler method with $h = 0.1$ to determine an approximation to the solution to the initial value problem $y' = y - x$, $y(0) = 1/2$ at $x = 1$. (starting from $n=1$ to $n=10$).

Question 5

- a) Thickness of a metal sheet (in mm) is measured repeatedly to obtain the following replicate data;

Experiment No	1	2	3	4	5	6
T (mm)	0.202	0.198	0.197	0.215	0.199	0.194
Experiment No	7	8	9	10	11	12
T (mm)	0.204	0.198	0.194	0.195	0.201	0.202

- What is the best estimate for the sheet thickness?
 - What is the variance of the distribution of errors with respect to the best value?
- b) It is given that the function $e^{-x^2/2}$ has a fourth derivative that is never greater than 3 in absolute value. i) use this fact to determine how many sub intervals are required for composite Simpsons rule to deliver an approximation to $\int_0^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$.

Question 6

- a) The Emf of a thermocouple was determined was determined at some temperature and tabulated as follows

$T^\circ(C)$	110	125	140	155
Emf (mv)	15.72	22.53	31.81	48.22

Obtain the connecting polynomial and hence estimate the emf of the thermocouple at $128^\circ C$.

- c) The function f is known to have a fourth derivative with the property that $f^{iv}(x) < 5$ for x between 1 and 5. Determine how many sub intervals are required so that the composite trapezium rule used to approximate $\int_1^5 f(x)dx$ incurs an error in it that is less than 0.005 and also for $\int_2^6 f(x)dx$ incurs an error that is less than 0.0005.